Adhithya G

B.E. Computer Science & Engineering (AI & Robotics)

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PROFILE

Motivated B.E. Computer Science (AI & Robotics) student with hands-on experience in robotics hardware, IoT, computer vision and machine learning. Has worked on a robot dog, an autonomous underwater waste-detection submarine and an AI-driven waste management framework. Brings strong communication, teamwork and event-organising skills, along with experience in AI video generation and video editing.

EDUCATION

B.E. Computer Science & Engineering (AI & Robotics)

2024 – 2028

Sathyabama Institute of Science & Technology, Chennai

CGPA: **9.0**

Higher Secondary Education

2015 – 2024

Chinmaya Vidyalaya Higher Secondary School, Vazhuthacaud

Class 10: **95%** | Class 12: **97%**

SKILLS

Technical

C | C++ | HTML | Python | Video Editing | AI Video Generation

Soft Skills

Communication | Teamwork | Time Management | Critical Thinking

WORK EXPERIENCE

Intern

ADB Labs & Orson Daily

Manufacturing · Video Generation · Marketing

Worked across manufacturing, AI video generation and marketing functions, gaining practical exposure to industry workflows and content production.

Event Organizer

College Event

Sathyabama Institute of Science & Technology

Planned and coordinated a college event, managing teamwork, scheduling and day-to-day execution.

SELECTED PROJECTS

Robot Dog (REX)

2026 – Present

Technologies: Python, Machine Learning, ROS 2, IoT, Embedded Systems, Robotics Hardware

- Explored and worked on AI-powered video generation for project demonstrations, promotional content and visual documentation.
- Contributed to robotics hardware development, including component integration, sensor interfacing and hardware-level implementation.
- Worked on IoT-based functionalities involving sensor connectivity, device communication and integration of electronic components.
- Worked on robot simulation using Gazebo, exploring virtual environments, testing movement and navigation behaviour, and supporting future ROS 2-based simulation and integration.

RoboSub (AQUA SWEEP)

2026 – Present

Technologies: Python, YOLO, Machine Learning, ESP32, ESP32-CAM, OpenCV, IoT, KiCad, 3D Modelling

- Worked on the development of a low-cost autonomous underwater robotic submarine for detecting recyclable waste and valuable objects in underwater environments.
- Contributed to 3D modelling and prototype development, including submarine hull design, thruster layout and structural component integration.
- Implemented AI-based object detection using YOLO and OpenCV with ESP32-CAM live video streaming to identify plastic waste and other underwater objects.
- Worked on IoT-based hardware integration involving camera connectivity, sensor interfacing, electronic components and wireless video communication.
- Explored future integration of a robotic gripper and collection basket for automated underwater waste collection.

AI-Driven Intelligent Waste Management

2026 – Present

Technologies: Python, Deep Learning, CNN, MobileNetV2, Machine Learning, Apache Hadoop, HDFS, Apache Spark, K-Means, Random Forest, scikit-learn

- Worked on developing an AI-driven waste classification framework using a pretrained CNN architecture (MobileNetV2) to categorize waste images into six classes.
- Explored image preprocessing techniques, including resizing, denoising, normalization and data augmentation to support waste classification.
- Worked on integrating classified waste records with Hadoop Distributed File System (HDFS) for distributed storage and large-scale data management.
- Applied K-Means clustering to identify waste generation patterns and group similar zones based on waste disposal behaviour.
- Worked on Random Forest-based forecasting to predict future waste generation volumes over a seven-day period.

LANGUAGES

English Tamil Malayalam Hindi

CERTIFICATIONS

- **Machine Learning Onramp** — MathWorks (Sept 2026)
- **Simulink Onramp** — MATLAB (Aug 2026)
- **Internship** — ADB Labs